

Nicholas Schuster

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EDUCATION

Lehigh University

M.S. Industrial Engineering and Operations Research

Bethlehem, Pennsylvania

August 2025 - December 2026

- Relevant Coursework: Optimization Models and Applications, Random Processes and Applications, Introduction to Machine Learning, Game Theory, Mathematics of Actuarial Science, Adv. Topics - Financial Risk Management

Lehigh University

B.S. Business Analytics and Finance

Bethlehem, Pennsylvania

August 2020 - May 2025

- Relevant Coursework: Predictive Analytics and Modeling, Stochastic Optimization, Deterministic Optimization, Introduction to Data Science, Database Management, Probability and Statistics, Linear Methods, Investments

WORK EXPERIENCE

Global Procurement Intern

May 2026 - Present

Carpenter Technology Corporation | Reading, PA

- Developed and executed Winshuttle scripts within SAP to update contract pricing records and purchase orders.
- Created a titanium ingot total cost of ownership (TCO) model to calculate capital expenditure and cost-savings.
- Migrated spend analytics systems into a centralized, queryable database format to improve ease of access.
- Redesigned supplier performance metrics to strengthen data-driven sourcing and vendor evaluations.

Continuous Improvement Intern

January 2026 - May 2026

Mack Trucks | Macungie, PA

- Designed and 3D printed fixtures to resolve fuel tank misalignment, minimizing related warranty claims by 40%.
- Piloted computer vision software to automate 3M analysis, building use cases for plant-wide line balancing.
- Optimized layout and material staging in the exhaust kitting area to cut down cycle times and improve traffic.

Procurement Analyst Intern

May 2025 - December 2025

Carpenter Technology Corporation | Reading, PA

- Leveraged Alteryx and Python to aggregate and report invoice data from multiple systems, improving visibility.
- Created a raw material pricing tracker to compare internal metal purchase prices against public exchange rates.
- Developed a cost-savings tracker that optimized process functions, streamlining department wide initiatives.

PROJECTS

Automobile Insurance Claims Frequency and Severity Modeling | *R, RStudio*

- Cleaned and pre-processed a historical dataset of 100,000+ auto insurance policies to identify key risk drivers.
- Developed a generalized linear model in R to predict both claims frequency as well as claims severity.
- Evaluated model performance using Gini coefficients and lift charts, providing ways to optimize policy pricing.

Catastrophic Risk Simulation and Capital Reserve Optimization | *Python, Monte Carlo Simulation, AMPL*

- Built a Python Monte Carlo model to simulate aggregate financial losses from catastrophic climate events.
- Formulated a linear program to determine the ideal capital reserve threshold subject to liquidity constraints.
- Conducted stress-testing and sensitivity analysis to evaluate inflation's impact on long-term claims payouts.

TECHNICAL SKILLS

- **Tools:** Microsoft Excel, Matlab, Minitab, AMPL, Oracle, RStudio, Power BI, Visual Studio Code
- **Programming Languages:** Python, R, SQL, VBA, Java
- **Spoken Languages:** English, Spanish
- **Awards:** Eagle Scout, Dean's List